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Agricultural machine suspension control system

Abstract

An agricultural machine includes a frame, a ground-engaging element, a suspension system that movably supports the frame relative to the ground-engaging element, wherein the suspension system is configured to apply, for a given displacement of the frame relative to the ground-engaging element, a force based on a force-to-displacement relationship. A control system is configured to receive an input indicative of an operational state of the agricultural machine during operation on a terrain, and automatically control the suspension system to adjust the force-to-displacement relationship of the suspension system based on the operational state.

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Background/Summary

FIELD OF THE DESCRIPTION

(1) The present description generally relates to suspension systems for mobile work machines. More specifically, but not by limitation, the present description relates to a control system that adjusts or tunes an agricultural machine suspension based on detecting a state of the machine or environment during operation of the machine.

BACKGROUND

(2) There are many different types of mobile work machines. Examples include, but are not limited to, agricultural machines, construction machines, turf management machines, forestry machines, among others. Some examples of agricultural machines include sprayers, tractors, harvesters, planters, seeders, to name a few. Many of these machines include a suspension system having components such as springs (e.g., air springs, etc.), shock absorbers, and other linkages that connect the machine to wheels, tracks, or other ground-engaging elements.

(3) The discussion above is merely provided for general background information and is not intended to be used as an aid in determining the scope of the claimed subject matter.

SUMMARY

(4) An agricultural machine includes a frame, a ground-engaging element, a suspension system that movably supports the frame relative to the ground-engaging element, wherein the suspension system is configured to apply, for a given displacement of the frame relative to the ground-engaging element, a force based on a force-to-displacement relationship. A control system is configured to receive an input indicative of an operational state of the agricultural machine during operation on a terrain, and automatically control the suspension system to adjust the force-to-displacement relationship of the suspension system based on the operational state.

(5) This Summary is provided to introduce a selection of concepts in a simplified form that is further described below in the Detailed Description. This Summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used as an aid in determining the scope of the claimed subject matter. The claimed subject matter is not limited to implementations that solve any or all disadvantages noted in the background.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 illustrates an example agricultural sprayer.

(2) FIG. 2 illustrates an example agricultural sprayer.

(3) FIG. 3 is a block diagram of one example of an agricultural spraying machine architecture.

- (4) FIG. 4 is a block diagram illustrating one example of a machine suspension controller.
- (5) FIG. 5 illustrates one example of a controllable section of a suspension subsystem.
- (6) FIG. 6 is a graph illustrating a spring rate curve of an example air spring.
- (7) FIG. 7 is a schematic diagram illustrating pneumatic circuitry of an example suspension subsystem.
- (8) FIGS. 8-1 and 8-2 (collectively referred to as FIG. 8) is a flow diagram illustrating an example operation to control a suspension system of an agricultural machine.
- (9) FIG. 9 is a block diagram showing one example of the architecture illustrated in FIG. 3, deployed in a remote server architecture.
- (10) FIGS. 10, 11, and 12 show examples of mobile devices that can be used in the architectures shown in the previous figures.
- (11) FIG. 13 is a block diagram showing one example of a computing environment that can be used in the architectures shown in the previous figures.

DETAILED DESCRIPTION

- (12) The present description generally relates to suspension systems for mobile work machines. More specifically, but not by limitation, the present description relates to a control system that adjusts or tunes an agricultural machine suspension based on detecting a state of the machine or environment during operation of the machine.
- (13) There are many different types of mobile work machines that utilize suspension systems. Examples include, but are not limited to, agricultural machines, construction machines, turf management machines, forestry machines, among others. Some examples of agricultural machines include sprayers, tractors, harvesters, planters, seeders, to name a few. For sake of illustration, but not by limitation, the present disclosure will be provided in the context of an agricultural sprayer or spraying machine. However, it will be understood that the present features can be utilized with other types of agricultural machines, as well as other types of mobile work machines.
- (14) FIG. 1 illustrates an agricultural spraying machine (or agricultural sprayer) **100**. Sprayer **100** includes a spraying system **102** having a tank **104** containing a liquid that is to be applied to field **106**. Tank **104** is fluidically coupled to spray nozzles **108** by a delivery system comprising a set of conduits. A fluid pump is configured to pump the liquid from tank **104** through the conduits through nozzles **108**. Spray nozzles **108** are coupled to, and spaced apart along, boom **110**. Boom **110** includes arms **112** and **114** which can articulate or pivot relative to a center frame **116**. Thus, arms **112** and **114** are movable between a storage or transport position and an extended or deployed position (shown in FIG. 1).
- (15) In the example illustrated in FIG. 1, sprayer **100** comprises a towed implement **118** that carries the spraying system, and is towed by a towing or support machine **120** (illustratively a tractor) having an operator compartment or cab **122**. Sprayer **100** includes a set of traction elements, such as wheels **124**. The traction elements can also be tracks, or other traction elements as well. It is noted that in other examples, sprayer **100** is self-propelled. That is, rather than being towed by a towing machine, the machine that carries the spraying system also includes propulsion and steering systems.
- (16) FIG. 2 illustrates one example of an agricultural sprayer **150** that is self-propelled. That is, sprayer **150** has an on-board spraying system **152**, that is carried on a machine frame **156** having an operator compartment **158**, a steering system **160** (e.g., wheels or other traction elements), and a propulsion system **162** (e.g., internal combustion engine).
- (17) FIG. 3 illustrates one example of an agricultural spraying machine architecture **200** having an agricultural spraying machine **202** configured to perform a spraying operation on an agricultural field. Examples of agricultural spraying machine **202** include, but are not limited to, sprayers **100** and **150** illustrated in FIGS. 1 and 2. Accordingly, machine **202** can comprise a towed implement or machine **202** can be self-propelled. FIG. 3 includes a dashed box **204** representing a towing machine, such as a tractor that is coupled to machine **202** through one or more links **206** (electrical,

mechanical, pneumatic, etc.).

(18) A control system **208** is configured to control components and systems of machine **202**. For instance, control system **208** includes a communication controller **210** configured to control a communication system **212** to communicate between components of machine **202** and/or with other systems, such as remote computing system **214** over a network **216**. Network **216** can be any of a wide variety of different types of networks such as the Internet, a cellular network, a local area network, a near field communication network, or any of a wide variety of other networks or combinations of networks or communication systems.

(19) A remote user **218** is shown interacting with remote computing system **214**. Remote computing system **214** can be a wide variety of different types of systems. For example, remote computing system **214** can be a remote server environment that is used by remote user **218**. Further, remote computing system **214** can be a mobile device, remote network, or a wide variety of other remote systems. Remote computing system **214** can include one or more processors or servers, a data store, and other items as well.

(20) Communication system **212** can include wireless communication logic, which can be substantially any wireless communication system that can be used by the systems and components of machine **202** to communicate information to other items, such as between computing system **208**, sensor(s) **220**, and controllable subsystems **222**. In one example, communication system **212** communicates over a controller area network (CAN) bus (or another network, such as an Ethernet network, etc.) to communicate information between those items. This information can include the various sensor signals and output signals generated by the sensor variables and/or sensed variables.

(21) Control system **208** is configured to control interfaces, such as operator interface(s) **224** that include input mechanisms configured to receive input from an operator **226** and output mechanisms that render outputs to operator **226**. The user input mechanisms can include mechanisms such as hardware buttons, switches, joysticks, keyboards, etc., as well as virtual mechanisms or actuators such as a virtual keyboard or actuators displayed on a touch sensitive screen. The output mechanisms can include display screens, speakers, etc.

(22) Sensor(s) **220** can include any of a number of different types of sensors. In the illustrated example, sensor(s) **220** include environmental sensor(s) **228** configured to sense characteristics of the environment in which machine **202** is operating, machine configuration or operational state sensor(s) **230** configured to sense configuration or operational characteristics of machine **202**, and can include other sensor(s) **232** as well.

(23) Example environmental sensor(s) **228** include weather sensors, such as wind speed and/or direction sensor(s) **234**, terrain sensor(s) **236**, and can include other sensor(s) **238**. Sensor(s) **234** are configured to sense a wind speed and/or direction on the field during operation of machine **202**. Terrain sensor(s) **236** are configured to sense characteristics of the field over which machine **202** is currently traveling or about to travel. For instance, sensor(s) **236** can detect the topography of the field to determine the degree of slope of various areas of the field, detect a boundary of the field, detect obstacles or other objects on the field (such as rocks, trees, etc.), among other things. In one example, one or more of sensor(s) **236** comprise an imaging system having image capture components configured to capture images and image processing components configured to process those images. In one example, image capture components include a stereo camera configured to capture video of the field being operated upon. An example stereo camera captures high definition video at thirty frames per second (FPS) with one hundred and ten degree wide-angle field of view. Of course, this is for sake of example only.

(24) Example machine configuration or operational state sensor(s) **230** include boom position and/or height sensor(s) **240**, boom and/or frame movement sensor(s) **242**, tread adjust sensor(s) **243**, machine weight sensor(s) **244**, material tank level sensor(s) **246**, machine attitude sensor(s) **248**, geographic position sensor(s) **250**, and can include other sensor(s) **252** as well.

(25) Sensor(s) **240** are configured to sense the current position and/or height of the boom of

machine **202**. Sensor(s) **242** are configured to sense movement of the boom. The sensors can be mounted on the boom, mounted on the frame that the boom is coupled to, or positioned otherwise. The sensors can include any suitable type of sensors including, but not limited to, accelerometers, gyroscopes, IMUs, to name a few.

(26) Tread adjust sensor(s) **243** are configured to detect machine tread width settings. This is discussed in further detail below. Machine weight sensor(s) **244** are configured to generate a sensor signal indicative of a weight of machine **202**, or a portion thereof. The signal can be utilized as an indication of load carried by the ground-engaging elements (e.g., wheels, tracks, etc.) of machine **202**. In one example, sensor(s) **244** include one or more material tank weight sensor(s) **254** configured to generate sensor signals indicative of a weight of material in material tank(s) on machine **202**. Thus, control system **208** can determine the weight of material tank(s) by directly sensing the load. Alternatively, or in addition, control system **208** can determine the weight of a material tank based on sensor signals from material tank level sensor **254**.

(27) Geographic position sensor(s) **250** include location sensor(s) **256**, heading/speed sensor(s) **258**, and can include other sensor(s) **260** as well. Location sensor(s) are configured to determine a geographic position of the machine **202** on the field. Location sensor(s) **256** can include, but are not limited to, a Global Navigation Satellite System (GNSS) receiver that receives signals from a GNSS satellite transmitter. Location sensor(s) **256** can also include a Real-Time Kinematic (RTK) component that is configured to enhance the precision of position data derived from the GNSS signal.

(28) Sensor(s) **258** are configured to determine a speed at which machine **202** is traversing the field during the spraying operation. Sensor(s) **258** can be configured to sense the movement of ground-engaging elements (e.g., wheels or tracks) and/or can utilize signals received from other sources, such as location sensor(s) **256**.

(29) Controllable subsystems **222** illustratively include a spraying subsystem **262**, boom position subsystem **264**, a propulsion subsystem **266**, a steering subsystem **268**, a suspension subsystem **270**, a tread adjustment subsystem **271**, and can include other subsystems **272** as well.

(30) Spraying subsystem **262** includes one or more pumps **274** configured to pump material from tanks **276** through conduits to nozzles **278** mounted on the boom. Spraying subsystem **262** can include other items **280** as well.

(31) Boom position subsystem **264** is configured to move the boom from a storage or transport position to a deployed position. In one example, boom position subsystem **264** includes actuators that are coupled to the boom and pivot the boom relative to a center or main frame.

(32) Propulsion subsystem **266** is configured to propel machine **202** across the field. Propulsion subsystem **266** can include a power source, such as an internal combustion engine, and a set of ground-engaging elements, such as wheels or tracks. Steering subsystem **268** configured to control the heading of the machine, by steering the ground-engaging elements.

(33) Suspension subsystem **270** is coupled to and supports the machine relative to the ground-engaging elements. Suspension subsystem **270** operably couples a frame of machine **202** to the ground-engaging elements (e.g., wheels, tracks, etc.). Suspension subsystem **270** includes one or more biasing elements, such as springs (e.g., mechanical springs, air springs, etc.), configured to apply a biasing force against the frame, which sets suspension system **270** at a particular stiffness and/or machine **202** at a particular ride height. That is, the biasing force of suspension subsystem **270** defines the stiffness of the ride experienced by operator **226** as the machine traverses the terrain. Suspension subsystem **270** can also include shock absorbers and other linkages that connects machine **202** to the ground-engaging elements.

(34) Tread adjustment subsystem **271** is configured to controllably adjust the tread width of machine **202**. For example, machine **202** can include movable axels that allow the width of the ground engaging elements (e.g., wheels, tracks, etc.) to be aligned to the width of the crop rows in the field being operated upon. In some instances, suspension performance can be significantly

impacted by tread width adjustments. Tread adjust sensor(s) **243** are configured to generate sensor signals indicative of the tread width (e.g., ground-engaging element position) by directly sensing the ground-engaging elements and/or detecting operation of tread adjustment subsystem **271**.

(35) As illustrated in FIG. **3**, control system **208** includes a machine velocity controller **282** configured to control the velocity of machine **202** by generating control signals for propulsion subsystem **266**. Control system **208** also includes a machine suspension controller **284** configured to control suspension subsystem **270**, discussed in further detail below, and can include other items **286** as well.

(36) Machine **202** includes a data store **288** configured to store data for use by machine **202**, such as field data **290** and/or suspension settings data **292**. Examples of field data **290** include field location data that identifies a location of the field to be operated upon by machine **202**, field shape information that identifies a shape of the field, and field topology data that defines the topology of the field. Examples of suspension settings data **292** include a selected or desired ride height and/or biasing force factors for suspension subsystem **270** correlated to particular terrain locations and/or operational settings or characteristics. For example, a user can set desired stiffness, damping, and/or ride-height settings for different in-field operations and on-road (or transport) operations. Data store **288** can store other items **293** as well.

(37) Machine **202** is also illustrated as including one or more processors or servers **294**, and can include other items **295** as well.

(38) As also illustrated in FIG. **3**, where a towing machine **204** tows agricultural spraying machine **202**, towing machine **204** can include some of the components discussed above with respect to machine **202**. For instance, towing machine **204** can include some or all of sensor(s) **220**, component(s) of control system **208**, and some or all of controllable subsystems **222**. Also, towing machine **204** can include a communication system **296** configured to communicate with communication system **212**, one or more processors or servers **297**, a data store **298**, and can include other items **299** as well.

(39) Machine suspension controller **284** of control system **208** is configured to control suspension subsystem **270** based on a configuration or operational state of machine **230**. Signals from sensors **220** indicate detected operational characteristics, such as, but not limited to, current and/or future machine speed, machine attitude or rotation, changes due to terrain characteristics, terrain objects, weather conditions, loading, speed, etc. As discussed in further detail below, the detected operational characteristics can be based on one or more of a priori data collected by control system **208** or in situ data generated based on sensor inputs, as well as user-defined parameters or other inputs.

(40) FIG. **4** is a block diagram illustrating one example of machine suspension controller **284** that includes suspension system control logic **302** configured to receive one or more inputs such as, but not limited to, a priori data **304**, in situ data **306**, and/or operator inputs **308** from operator **226** through operator interfaces **224**. A priori data **304** can be geo-referenced to locations in the field (or other terrain) upon which machine **202** is operating and can represent any of a variety of different operational states, such as machine configurations, commanded movements, environmental conditions, etc. This is discussed in further detail below. Illustratively, a priori data **304** is generated prior to the current operation of machine **202**. The data can be generated by machine **202**, or another machine or system (e.g., system **214**). One example includes a terrain or field map that identifies field characteristics, such as slope, soil condition, to name a few. Also, received map data can identify a projected or planned machine path, etc. For example, map data can identify headland turns in the field for the current machine operation. In another example, map data can identify turns along a roadway being traveled by the machine.

(41) In situ data **306** includes data generated based on sensor signals obtained during the current operation of machine **202**. For example, this data is obtained concurrently with operation of machine **202** based on sensor signals generated by sensors **220**. The in situ data **306** can represent

any of a variety of operational states related to the current operation of machine **202**. Examples include environmental characteristics, such as weather conditions, terrain conditions or topology, to name a few. Also, the in situ data can represent a configuration of machine **202**, such as configurations of controllable subsystems **222**. One example includes an indication as to whether the spray booms of spraying subsystem **262** are in a deployed position or a stowed/transport position. Further, the in situ data can represent a speed or heading of machine **202**, machine attitude, material levels, machine weight, etc. Operator inputs **308** can indicate suspension settings provided by operator **226** through operator interfaces **224**.

(42) Controller **284** includes suspension settings data generator logic **310**, suspension settings data correlation logic **312**, one or more processors **314**, and can include other items **316** as well.

Suspension settings data generation logic **310** is configured to generate suspension settings data **318** based on the inputs received by controller **284**. For example, as discussed in further detail below, suspension settings can define a particular force-to-displacement relationship (e.g., the spring rate or constant, or stiffness) of suspension subsystem **270**.

(43) At this point, it may be worth noting that suspension subsystem **270** can have a number of independently controllable sections. In the case of machine **202** having a pair of front ground-engaging elements (e.g., wheels, tracks, etc.) and a pair of rear ground-engaging elements, suspension subsystem **270** can have a single controllable section that jointly controls the spring rate or stiffness of the suspension coupling all four ground-engaging elements. In another example, suspension subsystem **270** includes a pair of controllable sections, that being a first or front controllable section for the pair of front-ground engaging elements and a second or rear controllable section for the rear ground-engaging elements. In yet another example, suspension subsystem **270** can have four controllable sections, each corresponding to one of the ground-engaging elements. In this way, the spring rate or stiffness of the suspension coupling each ground-engaging element is independently controllable of the other ground-engaging elements. In either case, each controllable section of suspension subsystem **270** corresponds to one or more of the ground-engaging elements of machine **202**.

(44) Logic **312** illustratively includes geo-referencing logic **320** configured to correlate or map suspension settings data **318** to particular geographic locations. In this way, the suspension settings data **318** can be utilized during subsequent operation of machine **202**, to control suspension subsystem **270** based on the suspension settings data **318**. For example, geo-referenced suspension settings data can indicate automatic and/or operator selected adjustments made to the suspension settings (i.e., for one or more controllable sections of suspension subsystem **270**) during a prior operation of machine **202** on a field. Suspension settings data **318** maps these suspension settings to the corresponding locations on the field at which they were used. Suspension settings data **318** can be stored in data store **288**, as indicated at suspension settings data **292**.

(45) Control logic **302** generates suspension control signals **322** to implement the suspension settings. The control signals **322** are provided to control suspension subsystem **270**. As noted above, suspension control signals **322** can correspond to multiple different independently controllable sections of suspension subsystem **270**. That is, a suspension control signal **322** can set different spring rate or stiffness settings for one or more of the ground-engaging elements.

(46) Accordingly, rather than using a predefined or present tuning (e.g., operating at a nominal vehicle ride height with a fixed suspension response or stiffness), suspension system **270** is dynamically adjustable during operation to accommodate changes in operational states (environmental and/or machine configuration). This can improve performance of the machine. For sake of illustration, the case of an example agricultural sprayer, precise application of the agricultural product (e.g., liquid fertilizer, herbicide, etc.) is important to achieve a desired level of effectiveness. If the product is unevenly applied, the product is wasted in areas of over-application, and areas of under-application can experience reduced yields. As the sprayer traverses across the field, disturbances such as changes in wheel height due to ground level changes, impact with

objects (such as rocks, trees, etc.), machine turns, and/or wind can induce movement in the boom, which can have undesirable effects on the boom position and orientation, and adversely affect the spraying performance. For example, the disturbances can change the vertical position of the nozzles, and thus the distance of the nozzles to the dispersal area in the field. Also, the dynamic adjustment can improve operator experience. A suspension that is tuned for a soft ride in rough field terrain may be too soft for during on-road transport, and vice versa. For instance, when traveling on-road, narrow winding roads or tight corners can result in a high degree of chassis roll which can be uncomfortable for operator **226**.

(47) FIG. **5** illustrates one example of a controllable section **350** of suspension subsystem **270**. As shown, a ground-engaging element **352** (illustratively a wheel) is mounted **9** to a hub driven by a shaft (not shown in FIG. **5**). The hub is carried on a pivotable assembly **354**, that is pivotably coupled to frame **356** at connection **358**. Rotation of assembly **354** is controlled by steering subsystem **268**.

(48) Section **350** is configured to apply a force, generally represented by arrow **360**, for a given displacement of frame **356** relative to element **352**. The applied force is a function of a force-to-displacement relationship defined by the configuration of section **350**. In the illustrated example, section **350** includes an air spring **362** operably positioned between the hub and frame **356**. An air spring refers to a component that is pneumatically driven to create a force in relation to deflection or displacement of the air spring. Examples include, but are not limited to, an air bag or air bellows. Movement of element **352** is constrained by rods or pistons **364** that are moveably disposed within cylinders **366** formed in assembly **354**. As discussed in further detail below, the force-to-displacement relationship of air spring **362** is controlled by increasing or decreasing the volume of air within the air spring **362** through control of a source of pressurized air, such as an air pump or compressor.

(49) FIG. **6** is a graph illustrating the spring rate curve of air spring **362**, in one example. As shown in FIG. **6**, air spring **362** has a non-linear relationship between force and displacement, or changes in the air spring height. To illustrate, for air spring heights that are lower (to the left on the x axis), air spring **362** has a higher spring rate, resulting in increased stiffness, whereas higher spring height (to the right on the x axis) have lower spring rates or a softer (less stiff) suspension performance.

(50) In accordance with one example, suspension subsystem **270** has a plurality of air spring height adjustment points between a maximum height (represented at reference numeral **400**) and a minimum height (represented at reference numeral **402**). Illustratively, two adjustment points (i.e., dual ride height adjustments) are provided. Of course, more than two adjustment points can be provided by machine suspension controller **284**.

(51) Machine suspension controller **284** is configured to adjust the pressure in air spring **362** to achieve a first nominal air spring height (represented at reference numeral **404**). Based on inputs during operation of machine **202**, machine suspension controller **284** is configured to reduce the pressure in air spring **362** to achieve a second nominal air spring height (represented at block reference numeral **406**). One example of a pneumatic system for achieving adjustments to suspension subsystem **270** is discussed below with respect to FIG. **7**. Briefly, however, the first nominal air spring height **404** resides at or near the mid-span position between the maximum height **404** and minimum height **402**. This, of course, is for sake of example only. Here, air spring **362** allows a range of travel above and below the first nominal height **404** when suspension subsystem **270** is configured at the first ride height set point. For example, air spring **362** allows four inches of travel above and below the first set point. Further, in the illustrated example, the second nominal height **406** is approximately two inches below the first set point **404**. Here, air spring **362** also allows a range of travel above and below the second set point, with an increased spring rate or stiffness.

(52) FIG. **7** is a schematic diagram illustrating pneumatic circuitry of suspension subsystem **270**, in one example. As shown, suspension subsystem **270** has a plurality of controllable sections **500**,

502, 504, and 506. For sake of discussion, section **500** will be described in further detail. It is noted that sections **502, 504, and 506** can include similar components as section **500**.

(53) Section **500** includes an air spring **508** (illustratively an air bag). A compressor **510** is driven by a motor **512** to provide pressurized airflow along a valve supply or inlet line **514**. A dual ride height suspension valve **516** is actuated by pilot pressure applied by actuation of a two-position, three-way valve **518**. This changes the suspension height control from the primary control valve to the secondary control valve. Under control from the second nominal height control valve, the suspension height is driven to the second nominal height position at which the slope of the spring rate curve is different than that at the first nominal height.

(54) FIGS. **8-1** and **8-2** (collectively referred to as FIG. **8**) is a flow diagram **600** illustrating an example operation to control a suspension system of an agricultural machine. For sake of illustration, but not by limitation, FIG. **8** will be described in the context of machine suspension controller **284** controlling suspension subsystem **270** of machine **202**, illustrated in FIG. **3**.

(55) At block **602**, control system **208** determines that machine **202** is operating on a particular terrain. This can include detecting a current geographic location at block **604** and determining a machine path at block **606**. For example, control system **208** can receive operator inputs that identify a field under operation along with a field plan for traversing the field during the operation.

(56) At block **608**, control system **208** identifies a current configuration of each controllable section of suspension subsystem **270**. As discussed above, suspension subsystem can include one or more controllable section, where each controllable section includes one or more ground-engaging elements.

(57) In one example, the current configuration can be determined based on manual operator input at block **610**. For example, operator **226** can set the stiffness of suspension subsystem **270**. In another example, a sensor input can be received from sensors **220** which indicate current settings of suspension subsystem **270**. The configuration indicates the current force-to-displacement relationship, or spring rate, of each controllable section. This is represented at block **614**. Alternatively, or in addition, the configuration can indicate the ride height at block **616**, damper and/or shock absorber settings at block **618**, and can indicate other configurations as well. This is represented at block **620**.

(58) At block **622**, one or more operational states of machine **202** are detected. In one example, the operational states indicate environmental conditions, which is represented at block **624** and can include receiving a priori data (block **626**) and/or in situ data (block **628**). The environmental conditions can indicate terrain characteristics, which is represented at block **630**. For example, the terrain characteristics can indicate a terrain topology map received by machine **202** or generated based on input from sensors **220**. Alternatively, or in addition, the terrain conditions can indicate soil conditions, such as moisture content, etc. Also, the environmental conditions can indicate weather conditions (block **632**), and can indicate other conditions as well. This is represented at block **634**.

(59) Also, the operational states detected at **622** can indicate machine configuration at block **636**. Again, this can be based on a priori data (block **638**) and/or in situ data (block **640**). The machine configuration can indicate the current and/or future attitude or rotation of machine **202** about three orthogonal axes (i.e., pitch, roll, yaw) as the machine traverses the terrain (block **642**). The machine configuration can also indicate the current or future speed of machine **202**. This is represented at block **644**. Also, the machine configuration can indicate particular configurations of subsystems **222**. This is represented at block **646**. For example, block **646** can indicate a current position of the spray boom, that is whether the spray boom is in a deployed or stowed/transport position. In another example, the machine configuration indicates the weight of machine **202** or portions thereof. For instance, a signal from material tank level sensor **246** indicates the amount of material in material tank **276**, which can be utilized to determine an estimated weight of machine **202**. In another example, the machine configuration indicates the configuration of tread adjustment

subsystem **271**. Of course, other machine configurations can be detected as well. This is represented at block **648**.

(60) At block **650**, an adjustment to the configuration of each controllable section of suspension subsystem **270** is determined. In one example, the determined adjustment is performed automatically, based on the detected operational states, detected at block **622**. This is represented at block **652**. In one example, the determined adjustment comprises a change to the force-to-displacement relationship, at block **654**. For example, the spring rate of an air spring in the controllable section is determined.

(61) Alternatively, or in addition, the adjustment can comprise a change to the ride height at block **656**. In the example above, a change in ride height is proportional, or otherwise related to, the change to the force-to-displacement relationship at block **654**. Also, the adjustment can comprise a change to damper and/or shock settings, at block **658**. In one example, adjustable dampers can be deployed in the controllable section. Control orifices on the damper can be adjusted to change the damping force applied by suspension subsystem **270**.

(62) The adjustment can be determined based on operator input, at block **660**. For example, operator **226** can enter desired suspension stiffness settings through operator interfaces **224**. The adjustment to the configuration can be determined in other ways as well. This is represented at block **662**.

(63) At block **664**, suspension settings data is generated by suspension settings data generator logic **310**, discussed above with respect to FIG. **4**. This data can be stored at block **666**, such as in data store **288**. Alternatively, or in addition, the suspension settings data can be sent to a remote system, such as remote system **214**. This is represented at block **668**. Of course, the suspension settings data can be generated and/or utilized in other ways as well. This is represented at block **670**.

(64) At block **672**, suspension system control logic **302** generates suspension control signals **322** to control suspension subsystem **270**. With respect to the example illustrated in FIG. **7**, control signals **322** control actuation of valve **518** to apply pilot pressure to dual ride height suspension valve **516**, which operates to change the suspension height control from the primary control valve to the secondary control valve. Under control from the secondary height control valve, the suspension height is driven to the secondary nominal height position at which the slope of the spring rate curve is different than at the primary nominal height. Upon removal of the pilot pressure at the suspension height control valves, the suspension is returned to the primary nominal ride height with the corresponding suspension performance. If, at block **674**, operation of machine **202** is continued, operation returns to block **604**.

(65) It can thus be seen that the present system provides a number of advantages. For example, but not by limitation, the present control system controls an agricultural machine suspension system based on the configuration or operational state of the machine, which can be based on geo-referenced a priori data as well as in situ data captured by on-board sensors of the machine. Using the detected operational states of the machine, automatic adjustments can be made to the force-to-displacement relationship of the suspension system. For example, the ride height of an air spring suspension is adjusted to change the stiffness of the suspension which can improve the performance of the machine operation as well as improve the operator experience.

(66) It will be noted that the above discussion has described a variety of different systems, components and/or logic. It will be appreciated that such systems, components and/or logic can be comprised of hardware items (such as processors and associated memory, or other processing components, some of which are described below) that perform the functions associated with those systems, components and/or logic. In addition, the systems, components and/or logic can be comprised of software that is loaded into a memory and is subsequently executed by a processor or server, or other computing component, as described below. The systems, components and/or logic can also be comprised of different combinations of hardware, software, firmware, etc., some examples of which are described below. These are only some examples of different structures that

can be used to form the systems, components and/or logic described above. Other structures can be used as well.

(67) The present discussion has mentioned processors, processing systems, controllers and/or servers. In one example, these can include computer processors with associated memory and timing circuitry, not separately shown. They are functional parts of the systems or devices to which they belong and are activated by, and facilitate the functionality of the other components or items in those systems.

(68) Also, a number of user interface displays have been discussed. They can take a wide variety of different forms and can have a wide variety of different user actuatable input mechanisms disposed thereon. For instance, the user actuatable input mechanisms can be text boxes, check boxes, icons, links, drop-down menus, search boxes, etc. They can also be actuated in a wide variety of different ways. For instance, they can be actuated using a point and click device (such as a track ball or mouse). They can be actuated using hardware buttons, switches, a joystick or keyboard, thumb switches or thumb pads, etc. They can also be actuated using a virtual keyboard or other virtual actuators. In addition, where the screen on which they are displayed is a touch sensitive screen, they can be actuated using touch gestures. Also, where the device that displays them has speech recognition components, they can be actuated using speech commands. Alternatively, or in addition, input devices are configured to detect gesture commands to control the machine.

(69) A number of data stores have also been discussed. It will be noted they can each be broken into multiple data stores. All can be local to the systems accessing them, all can be remote, or some can be local while others are remote. All of these configurations are contemplated herein.

(70) Also, the figures show a number of blocks with functionality ascribed to each block. It will be noted that fewer blocks can be used so the functionality is performed by fewer components. Also, more blocks can be used with the functionality distributed among more components.

(71) FIG. 9 is a block diagram of one example of the agricultural spraying machine architecture, shown in FIG. 3, where agricultural machine 202 communicates with elements in a remote server architecture 700. In an example, remote server architecture 700 can provide computation, software, data access, and storage services that do not require end-user knowledge of the physical location or configuration of the system that delivers the services. In various examples, remote servers can deliver the services over a wide area network, such as the internet, using appropriate protocols. For instance, remote servers can deliver applications over a wide area network and they can be accessed through a web browser or any other computing component. Software or components shown in FIG. 3 as well as the corresponding data, can be stored on servers at a remote location. The computing resources in a remote server environment can be consolidated at a remote data center location or they can be dispersed. Remote server infrastructures can deliver services through shared data centers, even though they appear as a single point of access for the user. Thus, the components and functions described herein can be provided from a remote server at a remote location using a remote server architecture. Alternatively, they can be provided from a conventional server, or they can be installed on client devices directly, or in other ways.

(72) In the example shown in FIG. 9, some items are similar to those shown in FIG. 3 and they are similarly numbered. FIG. 9 specifically shows that system 208 and data store 288 can be located at a remote server location 702. Therefore, agricultural machine 202 accesses those systems through remote server location 702.

(73) FIG. 9 also depicts another example of a remote server architecture. FIG. 9 shows that it is also contemplated that some elements of FIG. 3 are disposed at remote server location 702 while others are not. By way of example, data store 288 can be disposed at a location separate from location 702, and accessed through the remote server at location 702. Alternatively, or in addition, system 208 can be disposed at location(s) separate from location 702, and accessed through the remote server at location 702.

(74) Regardless of where they are located, they can be accessed directly by agricultural machine

202, through a network (either a wide area network or a local area network), they can be hosted at a remote site by a service, or they can be provided as a service, or accessed by a connection service that resides in a remote location. Also, the data can be stored in substantially any location and intermittently accessed by, or forwarded to, interested parties. For instance, physical carriers can be used instead of, or in addition to, electromagnetic wave carriers. In such an example, where cell coverage is poor or nonexistent, another mobile machine (such as a fuel truck) can have an automated information collection system. As the agricultural machine comes close to the fuel truck for fueling, the system automatically collects the information from the machine or transfers information to the machine using any type of ad-hoc wireless connection. The collected information can then be forwarded to the main network as the fuel truck reaches a location where there is cellular coverage (or other wireless coverage). For instance, the fuel truck may enter a covered location when traveling to fuel other machines or when at a main fuel storage location. All of these architectures are contemplated herein. Further, the information can be stored on the agricultural machine until the agricultural machine enters a covered location. The agricultural machine, itself, can then send and receive the information to/from the main network.

(75) It will also be noted that the elements of **FIG. 3**, or portions of them, can be disposed on a wide variety of different devices. Some of those devices include servers, desktop computers, laptop computers, tablet computers, or other mobile devices, such as palm top computers, cell phones, smart phones, multimedia players, personal digital assistants, etc.

(76) **FIG. 10** is a simplified block diagram of one illustrative example of a handheld or mobile computing device that can be used as a user's or client's handheld device **16**, in which the present system (or parts of it) can be deployed. For instance, a mobile device can be deployed in the operator compartment of agricultural machine **202** or as remote system **214**. **FIGS. 11-12** are examples of handheld or mobile devices.

(77) **FIG. 10** provides a general block diagram of the components of a client device **16** that can run some components shown in **FIG. 3**, that interacts with them, or both. In the device **16**, a communications link **13** is provided that allows the handheld device to communicate with other computing devices and under some embodiments provides a channel for receiving information automatically, such as by scanning. Examples of communications link **13** include allowing communication through one or more communication protocols, such as wireless services used to provide cellular access to a network, as well as protocols that provide local wireless connections to networks.

(78) In other examples, applications can be received on a removable Secure Digital (SD) card that is connected to an interface **15**. Interface **15** and communication links **13** communicate with a processor **17** (which can also embody processors or servers from previous 9 **FIGS.**) along a bus **19** that is also connected to memory **21** and input/output (I/O) components **23**, as well as clock **25** and location system **27**.

(79) I/O components **23**, in one example, are provided to facilitate input and output operations. I/O components **23** for various embodiments of the device **16** can include input components such as buttons, touch sensors, optical sensors, microphones, touch screens, proximity sensors, accelerometers, orientation sensors and output components such as a display device, a speaker, and or a printer port. Other I/O components **23** can be used as well.

(80) Clock **25** illustratively comprises a real time clock component that outputs a time and date. Clock **25** can also, illustratively, provide timing functions for processor **17**.

(81) Location system **27** illustratively includes a component that outputs a current geographical location of device **16**. This can include, for instance, a global positioning system (GPS) receiver, a LORAN system, a dead reckoning system, a cellular triangulation system, or other positioning system. Location system **27** can also include, for example, mapping software or navigation software that generates desired maps, navigation routes and other geographic functions.

(82) Memory **21** stores operating system **29**, network settings **31**, applications **33**, application

configuration settings **35**, data store **37**, communication drivers **39**, and communication configuration settings **41**. Memory **21** can include all types of tangible volatile and non-volatile computer-readable memory devices. Memory **21** can also include computer storage media (described below). Memory **21** stores computer readable instructions that, when executed by processor **17**, cause the processor to perform computer-implemented steps or functions according to the instructions. Processor **17** can be activated by other components to facilitate their functionality as well.

(83) FIG. **11** shows one example in which device **16** is a tablet computer **750**. In FIG. **11**, computer **750** is shown with user interface display screen **752**. Screen **752** can be a touch screen or a pen-enabled interface that receives inputs from a pen or stylus. Screen **752** can also use an on-screen virtual keyboard. Of course, screen **752** might also be attached to a keyboard or other user input device through a suitable attachment mechanism, such as a wireless link or USB port, for instance. Computer **750** can also illustratively receive voice inputs as well.

(84) FIG. **12** shows that the device can be a smart phone **71**. Smart phone **71** has a touch sensitive display **73** that displays icons or tiles or other user input mechanisms **75**. Mechanisms **75** can be used by a user to run applications, make calls, perform data transfer operations, etc. In general, smart phone **71** is built on a mobile operating system and offers more advanced computing capability and connectivity than a feature phone.

(85) Note that other forms of the devices **16** are possible.

(86) FIG. **13** is one example of a computing environment in which elements of FIG. **3**, or parts of it, (for example) can be deployed. With reference to FIG. **13**, an example system for implementing some embodiments includes a computing device in the form of a computer **810**. Components of computer **810** may include, but are not limited to, a processing unit **820** (which can comprise processors or servers from previous FIGS.), a system memory **830**, and a system bus **821** that couples various system components including the system memory to the processing unit **820**. The system bus **821** may be any of several types of bus structures including a memory bus or memory controller, a peripheral bus, and a local bus using any of a variety of bus architectures. Memory and programs described with respect to FIG. **3** can be deployed in corresponding portions of FIG. **13**.

(87) Computer **810** typically includes a variety of computer readable media. Computer readable media can be any available media that can be accessed by computer **810** and includes both volatile and nonvolatile media, removable and non-removable media. By way of example, and not limitation, computer readable media may comprise computer storage media and communication media. Computer storage media is different from, and does not include, a modulated data signal or carrier wave. Computer storage media includes hardware storage media including both volatile and nonvolatile, removable and non-removable media implemented in any method or technology for storage of information such as computer readable instructions, data structures, program modules or other data. Computer storage media includes, but is not limited to, RAM, ROM, EEPROM, flash memory or other memory technology, CD-ROM, digital versatile disks (DVD) or other optical disk storage, magnetic cassettes, magnetic tape, magnetic disk storage or other magnetic storage devices, or any other medium which can be used to store the desired information and which can be accessed by computer **810**. Communication media may embody computer readable instructions, data structures, program modules or other data in a transport mechanism and includes any information delivery media. The term “modulated data signal” means a signal that has one or more of its characteristics set or changed in such a manner as to encode information in the signal.

(88) The system memory **830** includes computer storage media in the form of volatile and/or nonvolatile memory such as read only memory (ROM) **831** and random access memory (RAM) **832**. A basic input/output system **833** (BIOS), containing the basic routines that help to transfer information between elements within computer **810**, such as during start-up, is typically stored in ROM **831**. RAM **832** typically contains data and/or program modules that are immediately accessible to and/or presently being operated on by processing unit **820**. By way of example, and

not limitation, FIG. 13 illustrates operating system **834**, application programs **835**, other program modules **836**, and program data **837**.

(89) The computer **810** may also include other removable/non-removable volatile/nonvolatile computer storage media. By way of example only, FIG. 13 illustrates a hard disk drive **841** that reads from or writes to non-removable, nonvolatile magnetic media, an optical disk drive **855**, and nonvolatile optical disk **856**. The hard disk drive **841** is typically connected to the system bus **821** through a non-removable memory interface such as interface **840**, and optical disk drive **855** is typically connected to the system bus **821** by a removable memory interface, such as interface **850**.

(90) Alternatively, or in addition, the functionality described herein can be performed, at least in part, by one or more hardware logic components. For example, and without limitation, illustrative types of hardware logic components that can be used include Field-programmable Gate Arrays (FPGAs), Application-specific Integrated Circuits (e.g., ASICs), Application-specific Standard Products (e.g., ASSPs), System-on-a-chip systems (SOCs), Complex Programmable Logic Devices (CPLDs), etc.

(91) The drives and their associated computer storage media discussed above and illustrated in FIG. 13, provide storage of computer readable instructions, data structures, program modules and other data for the computer **810**. In FIG. 13, for example, hard disk drive **841** is illustrated as storing operating system **844**, application programs **845**, other program modules **846**, and program data **847**. Note that these components can either be the same as or different from operating system **834**, application programs **835**, other program modules **836**, and program data **837**.

(92) A user may enter commands and information into the computer **810** through input devices such as a keyboard **862**, a microphone **863**, and a pointing device **861**, such as a mouse, trackball or touch pad. Other input devices (not shown) may include a joystick, game pad, satellite dish, scanner, or the like. These and other input devices are often connected to the processing unit **820** through a user input interface **860** that is coupled to the system bus, but may be connected by other interface and bus structures. A visual display **891** or other type of display device is also connected to the system bus **821** via an interface, such as a video interface **890**. In addition to the monitor, computers may also include other peripheral output devices such as speakers **897** and printer **896**, which may be connected through an output peripheral interface **895**.

(93) The computer **810** is operated in a networked environment using logical connections (such as a local area network-LAN, or wide area network—WAN or a controller area network—CAN) to one or more remote computers, such as a remote computer **880**.

(94) When used in a LAN networking environment, the computer **810** is connected to the LAN **871** through a network interface or adapter **870**. When used in a WAN networking environment, the computer **810** typically includes a modem **872** or other means for establishing communications over the WAN **873**, such as the Internet. In a networked environment, program modules may be stored in a remote memory storage device. FIG. 13 illustrates, for example, that remote application programs **885** can reside on remote computer **880**.

(95) It should also be noted that the different examples described herein can be combined in different ways. That is, parts of one or more examples can be combined with parts of one or more other examples. All of this is contemplated herein.

(96) Example 1 is an agricultural machine comprising: a frame; a ground-engaging element; a suspension system that movably supports the frame relative to the ground-engaging element, wherein the suspension system is configured to apply, for a given displacement of the frame relative to the ground-engaging element, a force based on a force-to-displacement relationship; and a control system configured to: receive an input indicative of an operational state of the agricultural machine during operation on a terrain; and automatically control the suspension system to adjust the force-to-displacement relationship of the suspension system based on the operational state.

(97) Example 2 is the agricultural machine of any or all previous examples, wherein the input comprises geo-referenced terrain data indicative of terrain characteristics of the terrain.

- (98) Example 3 is the agricultural machine of any or all previous examples, wherein the agricultural machine comprises an agricultural sprayer including a spraying system having a material tank configured to store a material to be sprayed and a set of spray nozzles.
- (99) Example 4 is the agricultural machine of any or all previous examples, wherein the operational state is indicative of an amount of the material in the material tank.
- (100) Example 5 is the agricultural machine of any or all previous examples, wherein the suspension system comprises an air spring configured to support the frame relative to the ground-engaging element, and the control system is configured to adjust a spring rate of the air spring based on the operational state.
- (101) Example 6 is the agricultural machine of any or all previous examples, wherein, when configured in the force-to-displacement relationship, the suspension system is positioned at a first ride height, and wherein the control system is configured to adjust the force-to-displacement relationship by changing the suspension system to a second ride height that is different than the first ride height.
- (102) Example 7 is the agricultural machine of any or all previous examples, wherein changing the suspension system to a second ride height comprises controlling a height control switching valve to select a control valve from one of a first control valve or a second control valve, wherein the selected control valve controls a flow of pressurized air to the air spring.
- (103) Example 8 is the agricultural machine of any or all previous examples, wherein the input comprises in situ data generated based on a sensor signal received during the operation of the agricultural machine.
- (104) Example 9 is the agricultural machine of any or all previous examples, wherein the sensor signal is generated by an environment sensor associated with the agricultural machine.
- (105) Example 10 is the agricultural machine of any or all previous examples, wherein the sensor signal represents a geographic location of the agricultural machine.
- (106) Example 11 is the agricultural machine of any or all previous examples, wherein the sensor signal represents an attitude of the machine.
- (107) Example 12 is the agricultural machine of any or all previous examples, wherein the sensor signal represents a configuration of a controllable subsystem of the machine.
- (108) Example 13 is the agricultural machine of any or all previous examples, wherein the input comprises a priori data generated prior to the operation of the agricultural machine on the terrain.
- (109) Example 14 is a method of controlling an agricultural machine, the method comprising: detecting an operational state of the agricultural machine during an operation on a terrain, the agricultural machine comprising a suspension system that movably supports a frame relative to a ground-engaging element and is configured to apply, for a given displacement of the frame relative to the ground-engaging element, a force based on a force-to-displacement relationship; determining an adjustment to the force-to-displacement relationship of the suspension based on the operational state; and controlling the suspension system based on the determined adjustment to the force-to-displacement relationship.
- (110) Example 15 is the method of any or all previous examples, wherein the input comprises geo-referenced terrain data indicative of terrain characteristics of the terrain.
- (111) Example 16 is the method of any or all previous examples, wherein the suspension system comprises an air spring configured to support the frame relative to the ground-engaging element, and controlling the suspension system comprises adjusting a spring rate of the air spring based on the operational state.
- (112) Example 17 is the method of any or all previous examples, wherein the sensor signal is generated by an environment sensor associated with the agricultural machine.
- (113) Example 18 is the method of any or all previous examples, wherein the sensor signal represents a configuration of a controllable subsystem of the machine.
- (114) Example 19 is a suspension system for an agricultural spraying machine, the suspension

system comprising: an air spring configured to support a frame of the agricultural machine relative to a ground engaging element by applying a force against the frame at a spring rate of the air spring; and a controller configured to: receive terrain data representing terrain characteristics; and automatically control the suspension system to adjust the spring rate of the air spring based on the terrain data.

(115) Example 20 is the suspension system of any or all previous examples, wherein the agricultural machine comprises an agricultural sprayer including a spraying system having a material tank configured to store a material to be sprayed and a set of spray nozzles, the air spring has a first spring rate at a first ride height, and the controller is configured to adjust the air spring to a second spring rate by changing the first ride height to a second ride height.

(116) Although the subject matter has been described in language specific to structural features and/or methodological acts, it is to be understood that the subject matter defined in the appended claims is not necessarily limited to the specific features or acts described above. Rather, the specific features and acts described above are disclosed as example forms of implementing the claims.

Claims

1. An agricultural machine comprising: a frame; a ground-engaging element; a suspension system that movably supports the frame relative to the ground-engaging element, wherein the suspension system comprises an air spring, having a flexible sidewall, configured to apply, for a given displacement of the frame relative to the ground-engaging element, a force based on a force-to-displacement relationship; and a control system configured to: receive an input indicative of an operational state of the agricultural machine during operation on a terrain; and automatically control the suspension system to adjust the force-to-displacement relationship by changing a configuration of the air spring, having the flexible sidewall, to adjust the force-to-displacement relationship of the air spring from a first force-to-displacement relationship to a second force-to-displacement relationship that is different than the first force-to-displacement relationship based on the operational state, wherein changing the configuration of the air spring comprises controlling a switching valve to select a selected control valve from one of a first control valve or a second control valve, wherein the selected control valve controls a flow of pressurized air to the air spring.
2. The agricultural machine of claim 1, wherein the input comprises geo-referenced terrain data indicative of terrain characteristics of the terrain.
3. The agricultural machine of claim 1, wherein the agricultural machine comprises an agricultural sprayer including a spraying system having a material tank configured to store a material to be sprayed and a set of spray nozzles and wherein the operational state is indicative of an amount of the material in the material tank.
4. The agricultural machine of claim 1, wherein the control system is configured to adjust a spring rate of the air spring based on the operational state.
5. The agricultural machine of claim 1, wherein the operational state comprises at least one of an environmental condition or a machine configuration.
6. The agricultural machine of claim 5, wherein the input comprises in situ data generated based on a sensor signal received during the operation of the agricultural machine.
7. The agricultural machine of claim 6, wherein the sensor signal is generated by an environment sensor associated with the agricultural machine.
8. The agricultural machine of claim 6, wherein the sensor signal represents at least one of an attitude of the agricultural machine or a geographic location of the agricultural machine.
9. The agricultural machine of claim 6, wherein the sensor signal represents a configuration of a controllable subsystem of the agricultural machine.
10. The agricultural machine of claim 1, wherein the input comprises a priori data generated prior to the operation of the agricultural machine on the terrain.

11. A method of controlling an agricultural machine, the method comprising: detecting an operational state of the agricultural machine during an operation on a terrain, the agricultural machine comprising a suspension system, including a first control valve, a second control valve, a switching valve, and an air spring, having a flexible sidewall, that movably supports a frame relative to a ground-engaging element and is configured to apply, for a given displacement of the frame relative to the ground-engaging element, a force based on a force-to-displacement relationship; determining an adjustment of the suspension system from a first force-to-displacement relationship to a second force-to-displacement relationship, that is different than the first force-to-displacement relationship, based on the operational state; and controlling the switching valve to select a selected control valve, from one of the first control valve or the second control valve, that controls a flow of pressurized air to the air spring, having the flexible sidewall, based on the determined adjustment.
 12. The method of claim 11, wherein receiving the operational state comprises receiving geo-referenced terrain data indicative of terrain characteristics of the terrain.
 13. The method of claim 11, wherein controlling the air spring comprises adjusting a spring rate of the air spring based on the operational state.
 14. The method of claim 11, wherein receiving the operational state comprises receiving a sensor signal generated by an environment sensor associated with the agricultural machine.
 15. The method of claim 11, wherein receiving the operational state comprises receiving a sensor signal representing a configuration of a controllable subsystem of the agricultural machine.
 16. A suspension system for an agricultural machine, the suspension system comprising: an air spring, having a flexible sidewall, configured to support a frame of the agricultural machine relative to a ground engaging element by applying a force against the frame at a spring rate of the air spring; and a controller configured to: receive terrain data representing terrain characteristics; and based on the terrain data, automatically control the air spring to adjust the spring rate of the air spring by controlling a switching valve to select a selected control valve from one of a first control valve or a second control valve, wherein the selected control valve controls a flow of pressurized air to the air spring.
 17. The suspension system of claim 16, wherein the agricultural machine comprises an agricultural sprayer including a spraying system having a material tank configured to store a material to be sprayed and a set of spray nozzles, the air spring has a first spring rate at a first ride height, and the controller is configured to adjust the air spring to a second spring rate by changing the first ride height to a second ride height.
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